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ME 1060: DYNAMICS

Lecture 3: Plane Kinetics of a Particle

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Overview

- Plane Kinetics of a Particle Applications in 1-d cases and 2 d cases
- Rectangular Cartesian coordinates
- n-t coordinates, and polar coordinates, curvilinear motion.

Kinetics of a Particle

Kinetics is a branch of dynamics that deals with the relationship between the change in motion of a body and the forces that cause this change.

The basis for kinetics is Newton's second law, which states that when:
an unbalanced force acts on a particle, the particle will accelerate in the direction of the force with a magnitude that is proportional to the force.

Kinetics of a Particle

Newton's second law of motion:

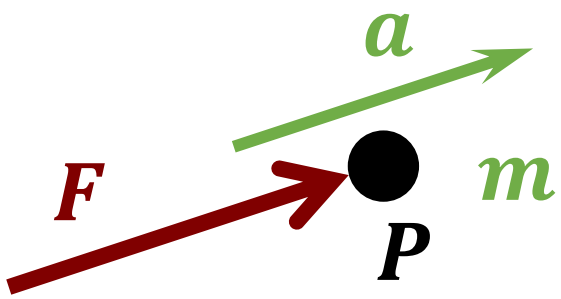
"A particle acted upon by an unbalanced force F experiences an acceleration *a* that has the same direction as the force and a magnitude that is directly proportional to the force"

or

"The unbalanced force acting on the particle is proportional to the time rate of change of the particle's linear momentum"

(Based on the assumption that time is an absolute quantity. This is valid for engineering problems concerned.)

Equation of Motion



M : mass of the particle P

F : unbalanced force

a : acceleration experienced

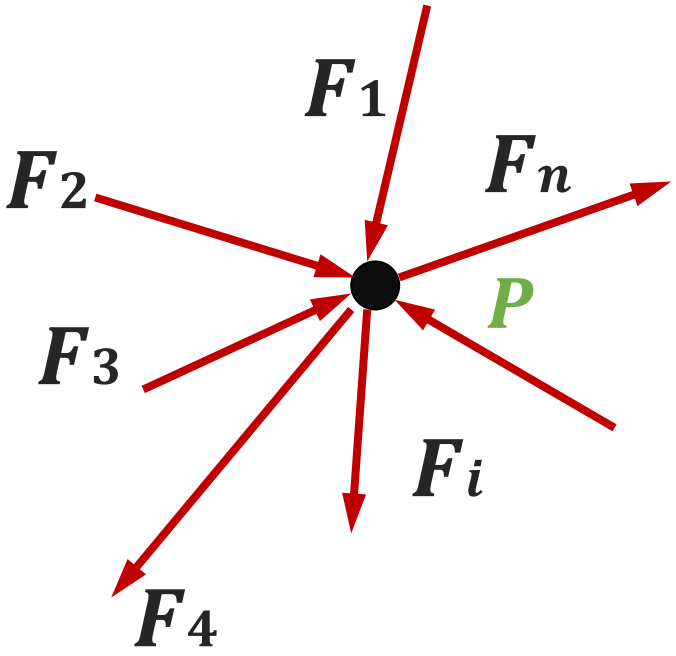
Newton's second law of motion mathematical form as

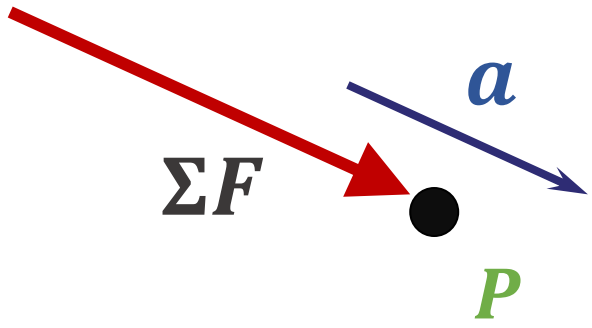
$$F = ma$$

referred to as the **Equation of Motion**

Kinetics of a Particle

When more than one force acts on a particle, the resultant force is determined by a vector summation of all the forces.





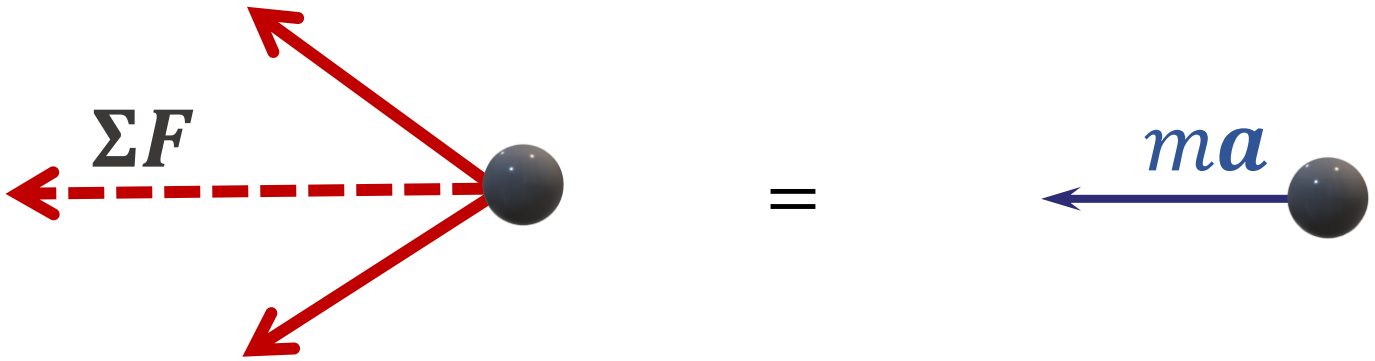
When a system of forces acting on a particle, (always a system of concurrent forces)

$\Sigma F = ma$,

where ΣF is the vector sum (or the resultant) of all the forces.

Free Body Diagram

We can graphically represent the magnitude and direction of each force acting on the particle by drawing the particle's free-body diagram,



Free – body Diagram

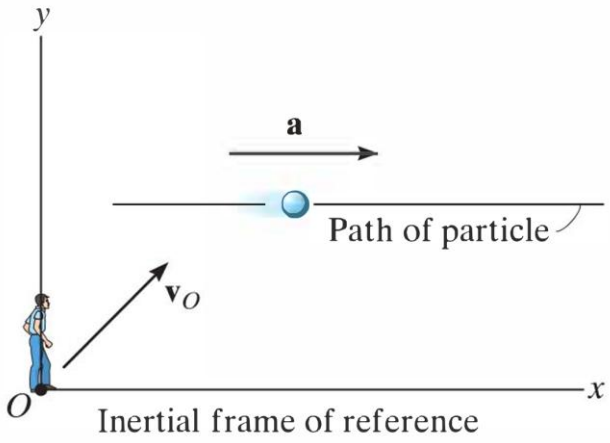
Kinetic Diagram

Inertial Reference Frame

For equations of motion, the acceleration of the particle has to be measured with respect to a reference frame that is either fixed or translates with a constant velocity.

In this way, the observer will not accelerate and measurements of the particle's acceleration will be the same from any reference of this type.

Such a frame of reference is commonly known as a **Newtonian** or **inertial reference frame**.



Kinetics of a Particle

D' Alembert principle

The equation of motion can be rewritten in the form

$$\Sigma F - ma = 0$$

-ma : inertia force or reaction vector.

Then the particle is in the state of equilibrium under the action of all the forces including this inertia force or reaction **(effective force).**

This is referred to as **Dynamic equilibrium**.

Equation of Motion for a System of Particles

System of particles

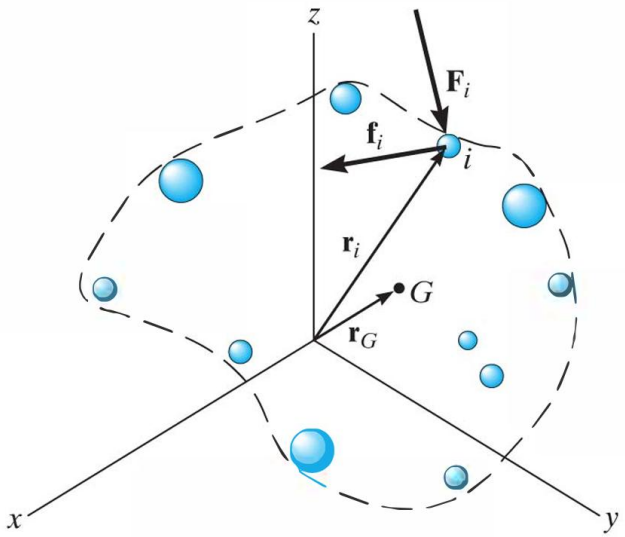
Equation of motion

$$\Sigma F_i = \Sigma m_i a_i \quad => \quad \Sigma F = m a_G$$

m = Total mass of the particles

a_G = The acceleration of the centre of mass of the system of particles

ΣF = The sum of the external forces acting on the system of particles



Inertial coordinate system

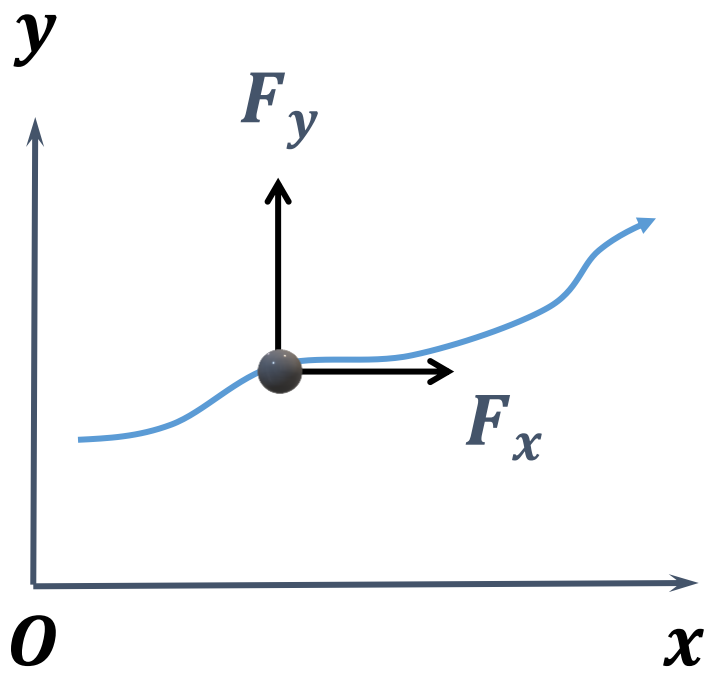
Hence, the sum of the external forces acting on the system of particles is equal to the total mass of the particles times the acceleration of its center of mass G.

Equations of Motion

2D rectangular Cartesian coordinates

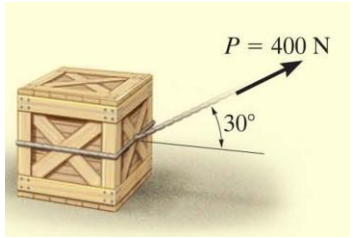
$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$



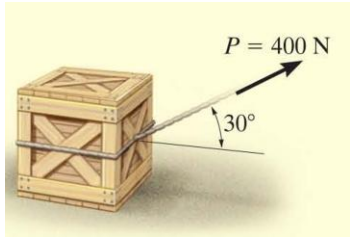
Example

The 50-kg crate shown in figure rests on a horizontal surface for which the coefficient of kinetic friction is $\mu_k = 0.3$. If the crate is subjected to a 400-N towing force as shown, determine the velocity of the crate in $3s$ starting from rest.



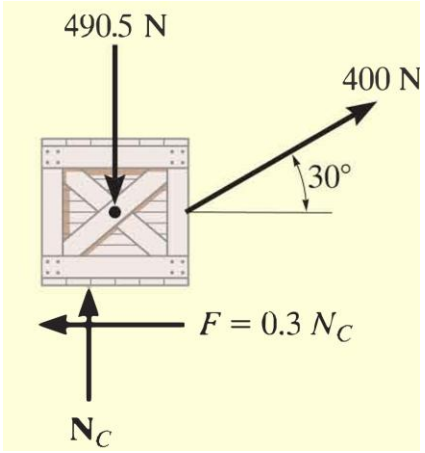
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Forces:

- The weight of the crate is $W = mg = 50\ kg\ (9.81\ ms^{-2}) = 490.5\ N.$
- Friction force $F = \mu N_c$ and acts to the left



Equation of Motion:

$\rightarrow \sum F_x = ma_x$

$400 \cos 30 - 0.3\ N_c = 50\ a$

$\uparrow \sum F_y = ma_y$

$N_c - 490.5 + 400 \sin 30 = 0$

$\boldsymbol{N_c = 290.5N}$

$\boldsymbol{a = 5.185\ ms^{-1}}$

Example

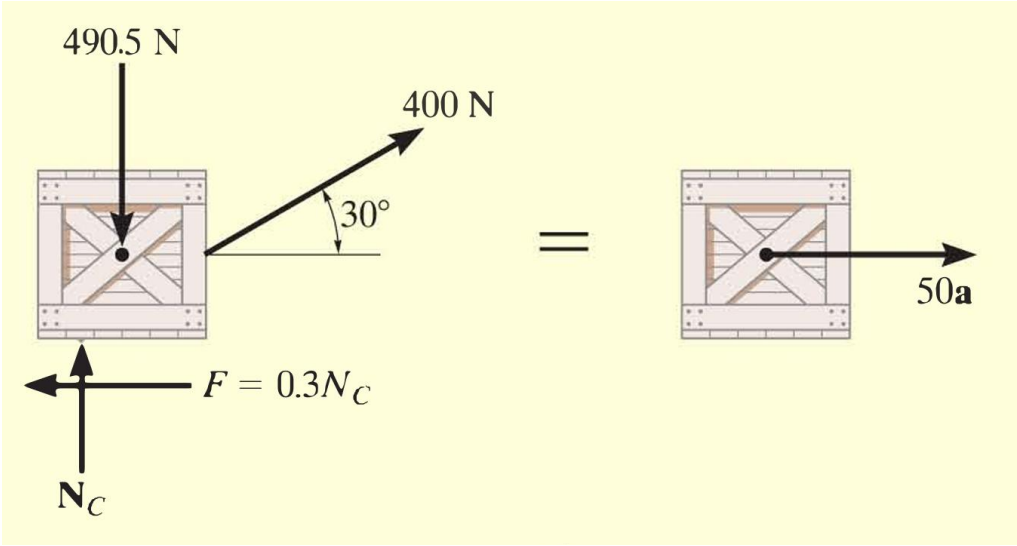
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Kinematics:

Notice that the acceleration is constant, since the applied force is constant.

Since the initial velocity is zero;

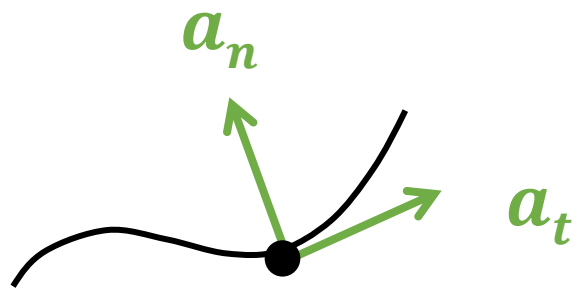
$$v = v_0 + a_c t = 0 + 5.185(3) = 15.6\, ms^{-1}\rightarrow$$



Equations of Motion

Normal and Tangential (N-T) coordinates

$$\Sigma F_t = ma_t$$
$$\Sigma F_n = ma_n$$



Inertial coordinate system

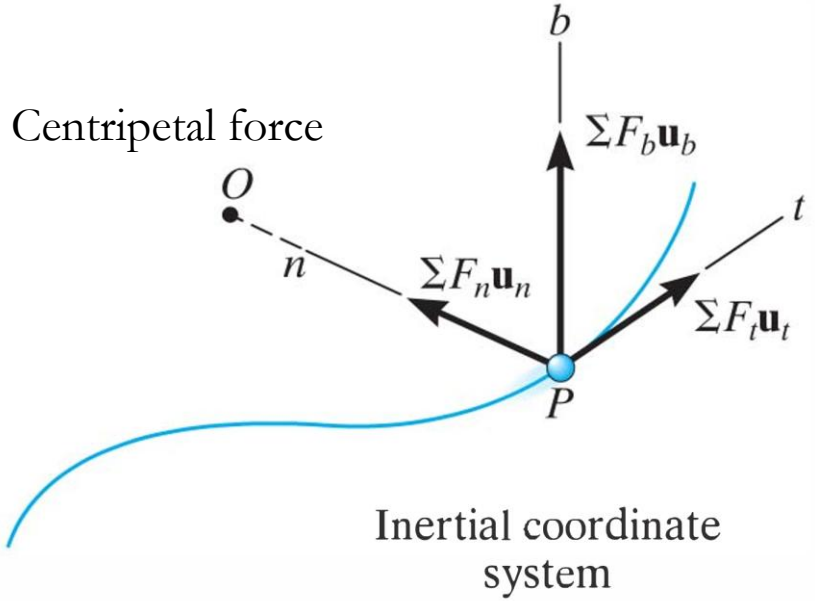
where,

$a_t = \frac{dv}{dt}$ represents the time rate of change in the magnitude of velocity.

$a_n = \frac{v^2}{\rho}$ represents the time rate of change in the velocity's direction.

If the path is defined as $y = f(x)$, the radius of curvature at the point where the particle is located can be obtained from

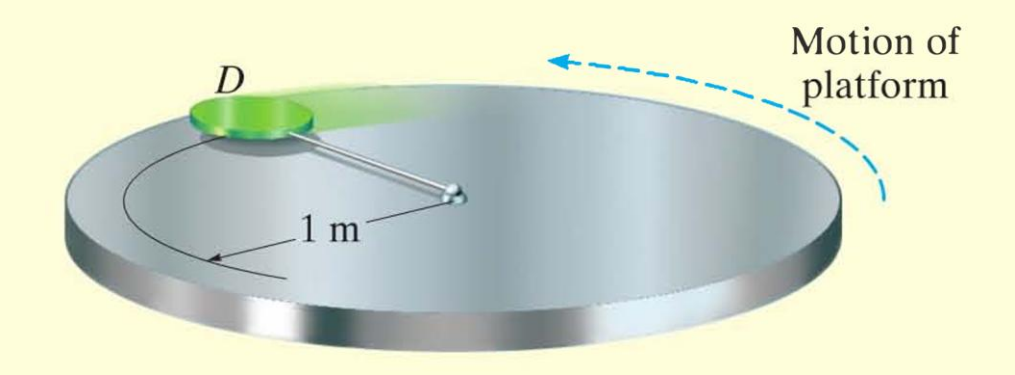
$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}.$$



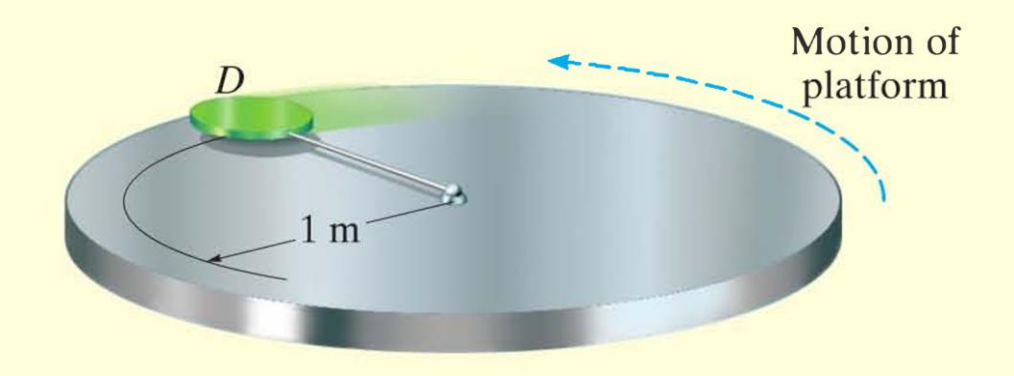
Inertial coordinate system

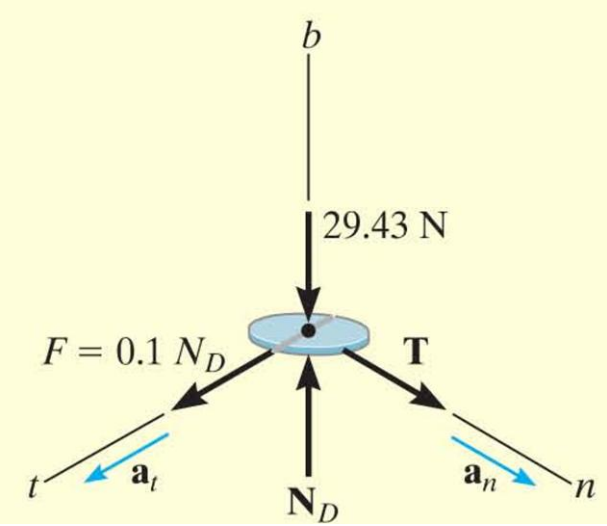
Example: Normal and Tangential (N-T) coordinates

The 3-kg disk D is attached to the end of a cord as shown in the figure. The other end of the cord is attached to a ball-and-socket joint located at the center of a platform. If the platform rotates rapidly, and the disk is placed on it and released from rest as shown, determine the time it takes for the disk to reach a speed great enough to break the cord. The maximum tension the cord can sustain is $100\ N$, and the coefficient of kinetic friction between the disk and the platform is $\mu_k = 0.1$.



Example: Normal and Tangential (N-T) coordinates





Free-Body Diagram.

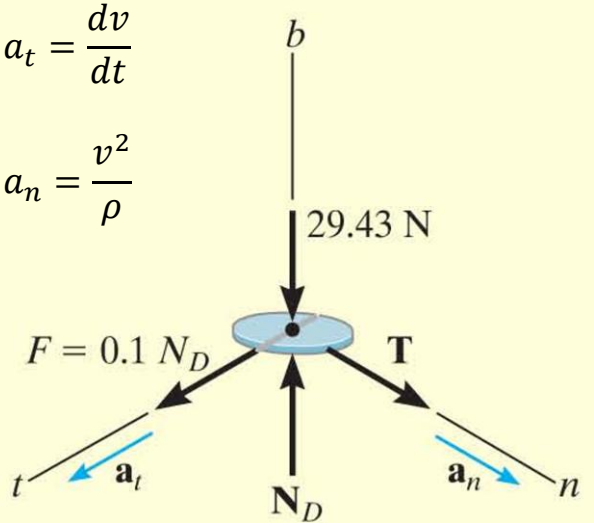
The frictional force has a magnitude $F = \mu_k N_D = 0.1\ N_D$ and a sense of direction that opposes the relative motion of the disk with respect to the platform.

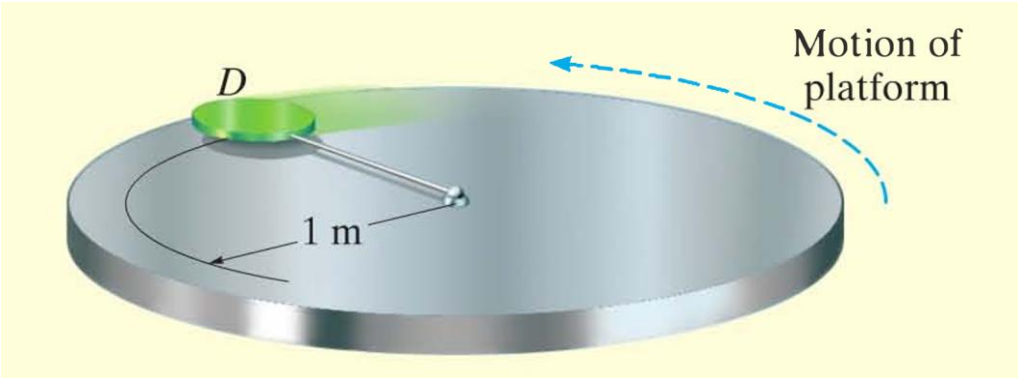
It is this force that gives the disk a tangential component of acceleration, causing v to increase, thereby causing T to increase until it reaches 100 N. The weight of the disk is $W = 3(9.81) = 29.43\ N$. Since a_n can be related to v, the unknowns are N_D , a_t, and v.

Example: Normal and Tangential (N-T) coordinates

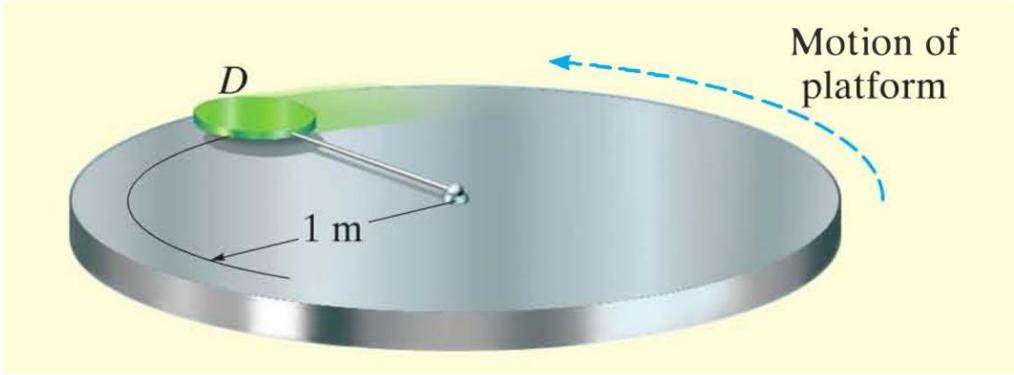
$$a_t = \frac{dv}{dt}$$

$$a_n = \frac{v^2}{\rho}$$





Example: Normal and Tangential (N-T) coordinates



Equation of Motion:

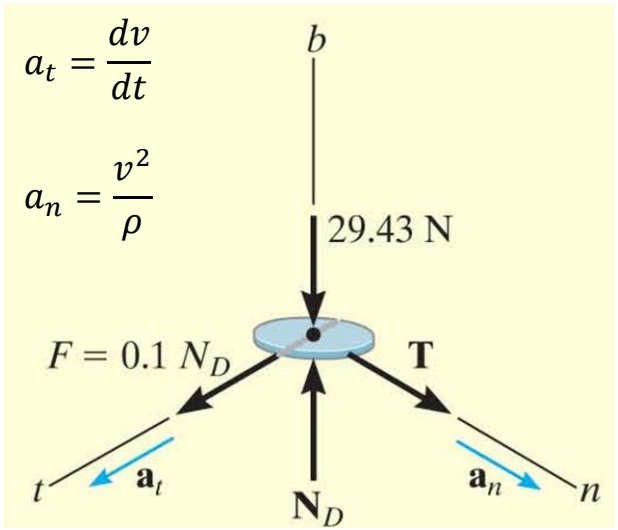
$$\sum F_n = ma_n; \quad T = 3\left(\frac{v^2}{1}\right)$$

$$\sum F_t = ma_t; \quad 0.1\,N_D = 3a_t$$

$$\sum F_b = 0; \quad N_D - 29.43 = 0$$

Solving the three equations when *T= 100 N*

$$N_D = 29.43\ N, \quad a_t = 0.981 ms^{-1} \quad v = 5.77\ ms^{-1}$$



Kinematics

$$v = v_0 + a_t t$$

$$5.77 = 0 + 0.981t$$

$$t = 5.89\ s$$

Equations of Motion

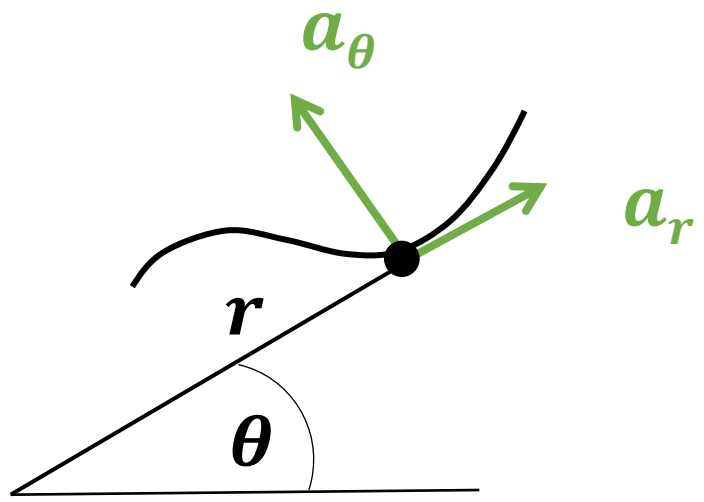
In 2D Polar coordinates

$$\Sigma F_r = ma_r$$
$$\Sigma F_\theta = ma_\theta$$

where,

$a_r = \ddot{r} - r\dot{\theta}^2$

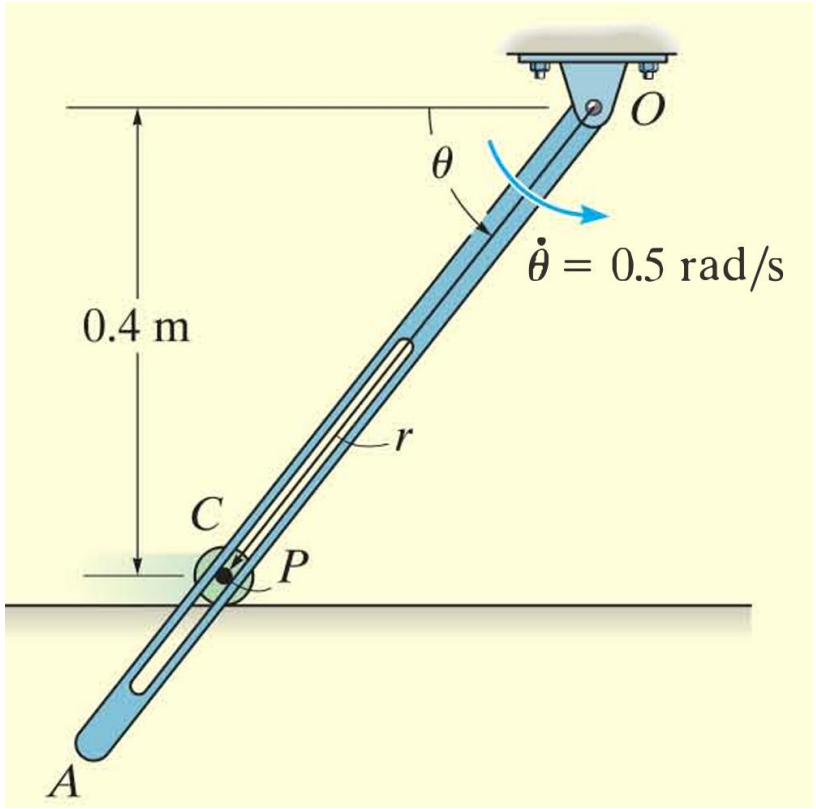
$a_\theta = r\ddot{\theta} + 2\dot{\theta}\dot{r}$



Inertial coordinate system

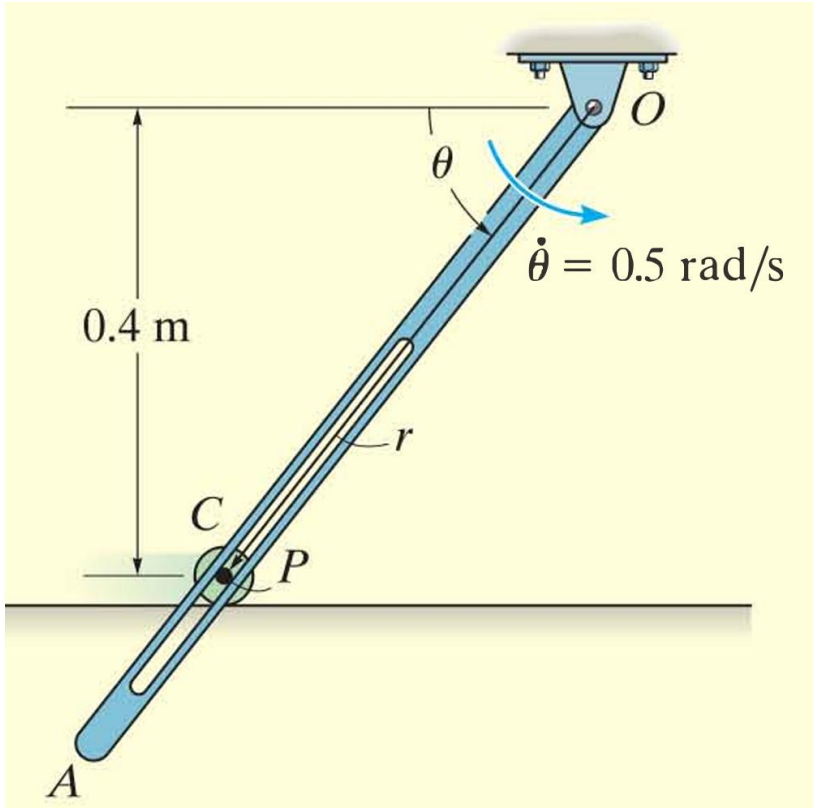
Example: In 2D Polar coordinates

The smooth 2-kg cylinder C has a pin P through its center which passes through the slot in arm OA. If the arm is forced to rotate in the vertical plane at a constant rate $\theta = 0.5\, rad/s$, determine the force that the arm exerts on the peg at the instant $\theta = 60°$.

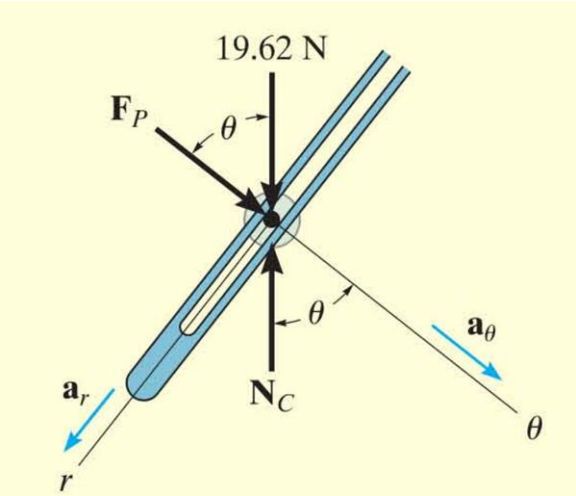


Example: In 2D Polar coordinates

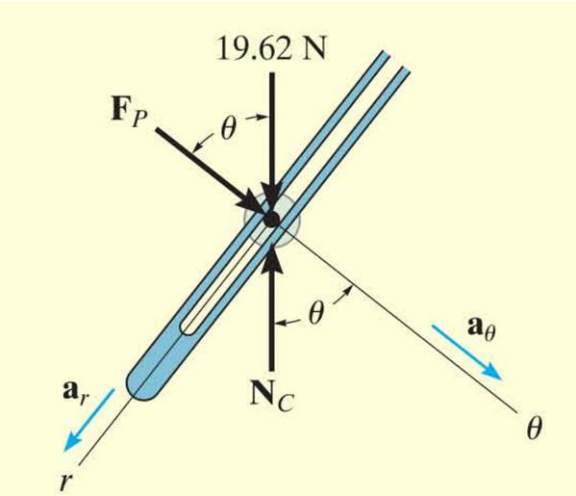
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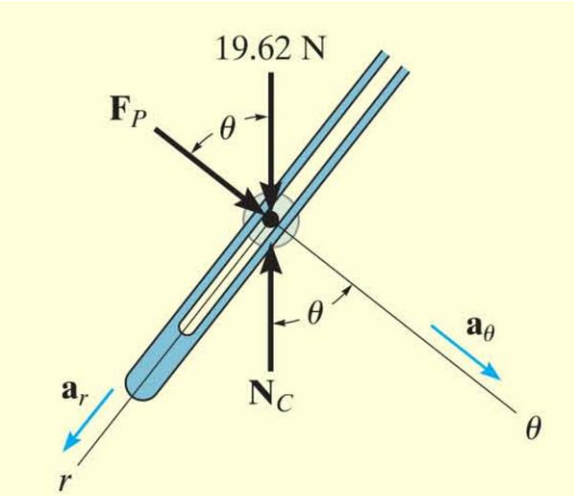
The force on the peg, F p, acts perpendicular to the slot in the arm. As usual, ar and ao are assumed to act in the directions of positive r and theta , respectively.



Example: In 2D Polar coordinates



Example: In 2D Polar coordinates



$\dot{\theta} = 0.5$

$\ddot{\theta} = 0$

Equation of Motion:

$\sum F_r = ma_r;\ \ 19.62\sin\theta - N_c\sin\theta = 2a_r$

$\sum F_\theta = ma_\theta; 19.62\cos\theta + F_p - N_c\cos\theta = 2a_\theta$

Kinematics

$$r = \frac{0.4}{\sin\theta} = 0.4\csc\theta$$

$\dot{r} = -0.4\ (csc\ \theta\ \cot\theta)\dot{\theta} = -0.2\ \csc\theta\ \cot\theta$

$\ddot{r} = -0.2(-\csc\theta\ \cot\theta)\ (\dot{\theta})\ \cot\theta - 0.2\ \csc\theta(-\csc^2\theta)\dot{\theta}$

$\ddot{r} = 0.1\ csc\theta\ (cot^2\theta + csc^2\theta)$

When $\theta = 60^o,\ r = 0.462, \dot{r} = -0.133, \ddot{r} = 0.192$

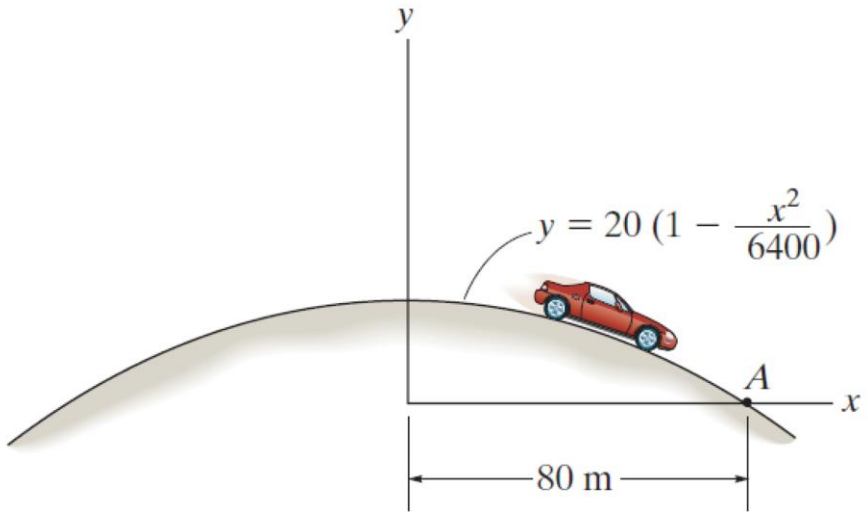
$a_r = \ddot{r} - r\dot{\theta}^2 = 0.192 - 0.462(0.5)^2 = 0.0770$

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-0.133)(0.5) = -0.133$

$N_c = 19.5\ N$ and $F_P = -0.356N$

Question

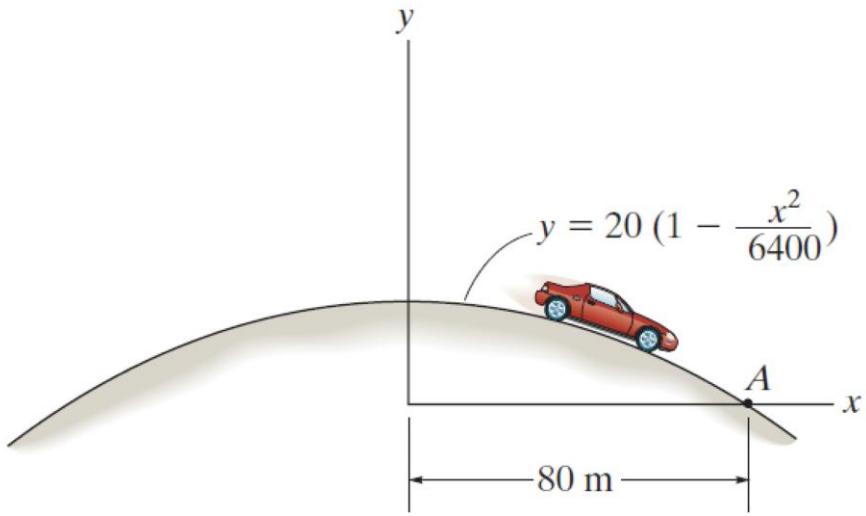
The 0.8-Mg car travels over the hill having the shape of a parabola. When the car is at point A, it is traveling at 9 m/s and increasing its speed at $3\ ms^{-2}$. Determine both the resultant normal force and resultant frictional force that all the wheels of the car exert on the road at this instant. Neglect the size of the car.

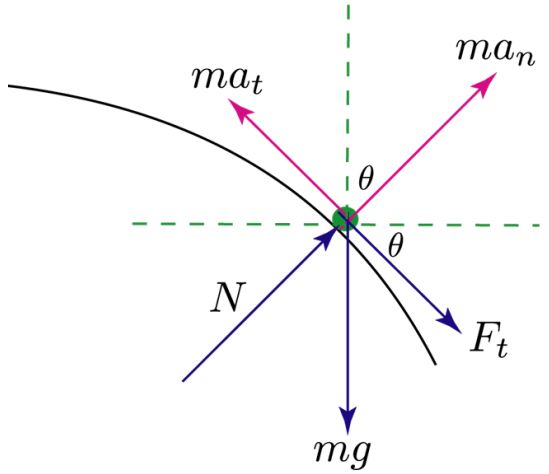


$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}.$$

Question

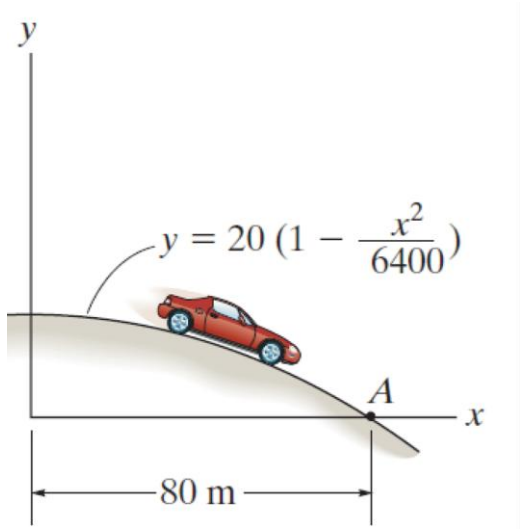
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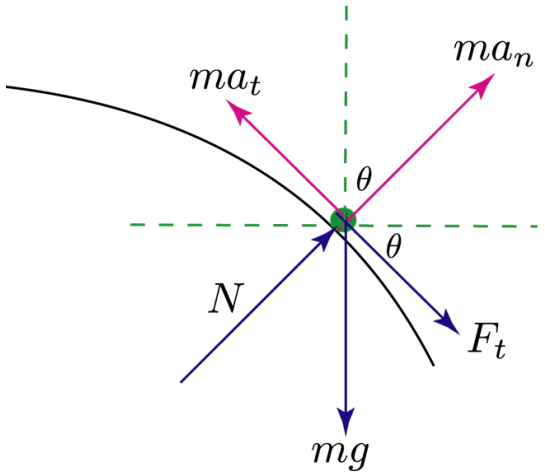




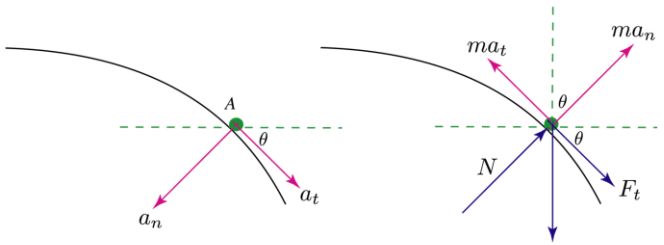
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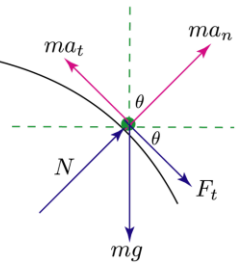
Question





Question





The car is taking a curvilinear motion and at A it has both tangential and normal accelerations.
Data given in the problem;

$$\begin{aligned} m &= 0.8 \text{ Mg} = 800 \text{ kg} \\ x &= 80 \text{ m} \\ v_t &= 9 \text{ m/s} \\ a_t &= 3 \text{ m/s}^2 \end{aligned}$$

At A, $x = 80$ m.

$$\frac{dy}{dx} = \frac{-40x}{6400} = -\frac{x}{160}$$

when $x = 80$ m,

$$\frac{dy}{dx} = -0.5$$

$$\frac{d^2y}{dx^2} = -\frac{1}{160}$$

$$\rho = \frac{[1 + (-0.5)^2]^{3/2}}{1/160} = 223.6 \text{ m}$$

$$a_n = \frac{v_t^2}{\rho} = 0.3622 \text{ m/s}^2$$

The angle θ can be found by;

$$\tan(-\theta) = \frac{dy}{dx} = -0.5$$

$$\theta = 26.57^\circ$$

Resolve forces;

$$\begin{aligned} \nearrow \quad & N + ma_n - mg\cos\theta = 0 \\ \searrow \quad & F_t + mg\sin\theta - ma_t = 0 \end{aligned}$$

$$\begin{aligned} N &= m(-a_n + g\cos\theta) = 800(-0.3622 + 9.81\cos(26.57^\circ) \\ F_t &= m(-g\sin\theta + a_t) = 800(-9.81\sin(26.57^\circ) + 3) \end{aligned}$$

$$\begin{aligned} N &= 6729 \text{ N} \\ F_t &= -1110 \text{ N} \end{aligned}$$